

Lab 10: Lab-ubu

PSTAT 100, Summer Session A 2025 with Ethan P. Marzban

MEMBER 1 (NetID 1) MEMBER 2 (NetID 2)
MEMBER 3 (NetID 3)

July 24, 2025

Required Packages

```
library(ottr)      # for checking test cases (i.e. autograding)
library(pander)    # for nicer-looking formatting of dataframe outputs
library(tidyverse) # for graphs, data wrangling, etc.
```

Logistical Details

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- This lab is due by **11:59pm on Wednesday, July 30, 2025**.
- Collaboration is allowed, and encouraged!
 - If you work in groups, list ALL of your group members' names and NetIDs (not Perm Numbers) in the appropriate spaces in the YAML header above.
 - Please delete any "MEMBER X" lines in the YAML header that are not needed.
 - No more than 3 people in a group, please.
- Ensure your Lab properly renders to a **.pdf**; non-**.pdf** submissions will not be graded and will receive a score of 0.
- Ensure all test cases pass (test cases that have passed will display a message stating "All tests passed!")

Lab Overview and Objectives

Welcome to another PSTAT 100 Lab! In this lab, we will cover the following:

- Clustering in R
- Missing Data

Part I: Clustering

Labubus are a moderately-sized plush doll that has completely taken the online market recently. If you'd like to read more about the history of Labubus, feel free to consult [This NPR Article](#).

For the purposes of this lab, we'll explore a simulated dataset pertaining to *Labubus*. Specifically, we are imagining that someone has collected 61 labubus and recorded the following attributes:

- The height of the *Labubu* (in centimeters)
- The weight of the *Labubu* (in grams)
- The RGB color specification of the *Labubu* (split across three columns; one for R, one for G, and one for B)

Additionally, it is known that each *Labubu* falls into one of six styles: Soymilk (SM), Lychee Berry (LB), Green Grape (GG), Sea Salt Coconut (SSC), Toffee (T), and Sesame Bean (SB) (names taken from [this](#) listing).

The full *Labubu* dataset can be found in the `labubu.csv` file, located in the `data/` subfolder.

```
labubu <- read.csv("data/labubu.csv")
```

! Question 1

Create a data frame called `labubu_num` that contains only the numerical columns from the `labubu` dataset.

Solution:

```
## replace this line with your code
```

Answer Check:

```
# DO NOT EDIT THIS LINE
invisible({check("tests/q1.R")})
```

Test q1 failed:

```
object 'labubu_num' not found
```

Let's see what happens if we try to visually classify based on only two variables.

! Question 2

Generate a scatterplot of the Labubus' heights (on the vertical axis) against their weights (on the horizontal) axis. Ensure your plot is presentation-quality. How many clusters do you see on the plot? **Don't** rely on the `type` column from the `labubu` dataframe - only use the plot you just generated!

Solution:

```
## replace this line with your code
```

Replace this line with your answer

Answer Check:

There is no autograder for this question; your TA will manually check that your answers are correct.

Hm, our number of clusters doesn't appear to agree with the number of different *Labubu* types we know to be present in the dataset. This *could* mean that two or more of the *Labubu* types are very similar to one another - however, it's dangerous to make that conclusion based solely on our work in Question 2 since we ignored several variables from the dataset. Let's remedy that!

! Question 3

Perform PCA on the `labubu_num` dataframe, taking care to *standardize* (not only mean-center) each column. Then, use this to generate a plot of the first two latent factors; how many clusters do you see on this PC plot?

Solution:

```
## replace this line with your code
```

Replace this line with your answer

Answer Check:

There is no autograder for this question; your TA will manually check that your answers are correct.

Wow- the clusters on this graph are pretty distinct! Nevertheless, let's run a *K*-means clustering algorithm on our PC-projected dataset. As a reminder, in **R** we use the `kmeans()` function to run the *K*-means clustering algorithm; take a look at the help file for this function before moving onto the next question on this lab.

! Question 4**Part (a)**

Pass the matrix of the first two PCs into a call to `kmeans()`, and assign the result to a variable called `kmeans_labubu`. Use 6 clusters. **Make sure to set your seed to 100 before beginning; otherwise the autograder may not work.**

Solution:

```
set.seed(100)
## replace this line with your code
```

Answer Check:

```
# DO NOT EDIT THIS LINE
invisible({check("tests/q4a.R")})
```

Test q4a failed:

object 'kmeans_labubu' not found

Part (b)

Re-do the PC plot from Question 3, now coloring based on the clusters identified by `kmeans()`.

```
## replace this line with your code
```

Answer Check:

There is no autograder for this question; your TA will manually check that your answers are correct.

Assuming you correctly completed the questions above, the following code should produce a cross-tabulation of each *Labubu*'s classification (as determined by *K*-means) and its true type (as classified by the `type` variable from the original `labubu` dataframe):

```
labubu %>%
  mutate(class = kmeans_labubu$cluster) %>%
  group_by(type, class) %>%
  summarise(count = n()) %>%
  pivot_wider(names_from = type,
              values_from = count,
              values_fill = 0) %>%
  pander()
```

```
Error in `mutate()` :
i In argument: `class = kmeans_labubu$cluster`.
```

Caused by error:

```
! object 'kmeans_labubu' not found
```

```
Error in if (tail(stdout, 1) == "") {: argument is of length zero
```

! Question 5

Has K -means classified along the different types, as encoded by the `type` variable in the original `labubu` dataframe? Are there any two *Labubu* types that are so similar to each other that they are indistinguishable (by K -means) from one another? How can you tell?

Solution:

Replace this line with your answer

Answer Check:

There is no autograder for this question; your TA will manually check that your answers are correct.

Something that is important to note is that there is a bit of randomness in the K -means algorithm. As such, let's see how this manifests itself in this particular dataset.

! Question 6

Re-run the K -means clustering algorithm, but this time **set your seed to 20 at the start**. Produce a plot like you did in Question 4(b), coloring points by the *new* classifications. Comment on any differences between this plot and the plot from 4(b) above.

```
set.seed(20)
## replace this line with your code
```

Replace this line with your answer

Answer Check:

There is no autograder for this question; your TA will manually check that your answers are correct.

So, there is some variability in the product of the K -means clustering algorithm. Again, recall that the K -means algorithm is only optimal in a local sense - in some cases (like with this particular simulated *Labubu* dataset), the final configurations may be dependent on the seed you set initially.

Submission Details

Congrats on finishing this PSTAT 100 lab! Please carry out the following steps:

i Submission Details

- 1) Check that all of your tables, plots, and code outputs are rendering correctly in your final .pdf.
- 2) Check that you passed all of the test cases (on questions that have autograders). You'll know that you passed all tests for a particular problem when you get the message "All tests passed!".
- 3) Submit **ONLY** your .pdf to Gradescope. Make sure to **match ALL pages to the ONE question on Gradescope**; failure to do so will incur a penalty of 0.1 points.